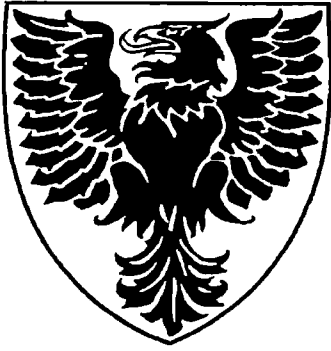




R. Brown

Dalhousie University

Department of Psychology

24th Annual In-House Convention

29 April-1 May 1998

Wednesday, April 29, 1998 (LSC 240)

- 1:30 **Introduction to the In-House Convention** —Richard Brown
- 2:00- 3:00 **Session 1.** Chair: Kelly Purdy
 T1) Bruce Dick
 T2) Myong Yoon
 T3) Lianne Stanford —
 T4) Renee Duquesnay
- 3:00- 3:30 **Break**
- 3:30- 5:00 **Session 2.** Chair: Mike Davies
 T5) Jeff Hancock
 T6) Kelly Shoemaker
 T7) Nadine Rossy
 T8) Fernando Larrea
 T9) Kim Good
 T10) Ben Tremblay

Thursday, April 30, 1998 (LSC 240)

- 8:45 - 10:00 **Session 3.** Chair: Heather Maxwell
 T11) Ryan D'Arcy
 T12) Chris Moore
 T13) Tara Fidler
 T14) William Brown
 T15) John Christie
- 10:00 - 10:30 **Coffee Break**
- 10:30 - 12:00 **Session 4.** Chair: Michelle Gregus
 T16) Jim Blackburn
 T17) Richard Brown —
 T18) Xiawei Song
 T19) Jean Saint-Aubin
 T20) Sherry Stewart
 T21) Boris Palameta
- 12:00 - 1:30 **Lunch Break and Posters (Lounge)**
- 1:30 - 3:00 **Session 5.** Chair: Randall Henderson
 T22) Ben Rusak
 T23) David Shore
 T24) Doug Symons
 T25) Cathy Forestell
 T26) Seth Bloom
 T27) Nancy Comeau
- 3:00- 3:30 **Coffee Break**
- 3:30- 5:00 **Keynote Speech**

Friday, May 1, 1998 (LSC 240)

8:30 **Beginning of the Conference Day**

8:45 - 10:00 **Session 6.** Chair: Lesley Varney

- T28) Alison Cole
- T29) Ray Klein
- T30) Anna Lee
- T31) Guy Gingras
- T32) Glenn Schellenberg

10:00 - 10:30 **Coffee Break**

10:30 - 12:00 **Session 7.** Chair: Patty Dill

- T33) Dennis Phillips
- T34) Susan Boehnke
- T35) Kazue Semba
- T36) Jason Emsley
- T37) Tim Bredy (Will be presented as a POSTER)
- T38) Elizabeth McLaughlin

12:00 - 1:30 **Lunch Break and Posters (Lounge)**

1:30 - 3:00 **Session 8.** Chair: Boyer Winters

- T39) Susan Jerrott
- T40) Sonia Greaven
- T41) Ivar Mendez
- T42) Veronique MacKenzie-Bourne
- T43) Marleen deGroot
- T44) Robin Patterson

3:00- 3:30 **Coffee Break**

3:30- 5:30 **Symposium "To Hebb and Beyond"**

by the Dalhousie University Psychology Proseminar Group

- Susan Boehnke
- Bruce Dick
- Jeff Hancock
- Ray Klein
- Nadine Rossy
- Kelly Shoemaker
- Lianne Stanford

There will be pizza and beer available in the Lounge after the symposium.

All presentations are listed in the name of the first author from the abstracts submitted.

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Welcome to the 1998 Psychology and Neuroscience In-House Convention

Welcome to the Twenty Fourth Annual Dalhousie University Department of Psychology In-House Convention. This year's meeting is being hosted by the students and faculty of our Graduate Program in Psychology. The program planning committee consists of the following individuals: Myong G. Yoon, David Rioux, William M. Brown, Kelly Shoemaker, Lianne Stanford, Guy Gingras, and Marleen de Groot. A special thanks to Suzanne King for efforts above and beyond the call of duty.

Friendly Reminder

Please wear your name-tag for all events associated with the convention. This will facilitate conversation and allow access to the lunches and the pizza banquet.

T1. The Effects of Pain on Event-Related Potentials

Bruce Dick

The importance of accurate pain assessment and measurement cannot be overstated. This presentation will briefly outline the progress in pain assessment leading up to the research to be presented. As a result of collaboration between the Pediatric Pain Lab and the Cognitive/Clinical Neuroscience Unit, research has investigated the utility of event-related potentials (ERPs) as a pain measurement tool. This work capitalizes on the cognitive relation between pain and distraction and is founded on an attentional resource allocation model. Distraction appears to channel attention away from pain stimuli, and the P300 ERP wave amplitude has been found to be related to attentional resource utilization. The proposed model therefore predicts that pain will act as a consumer of attentional resources and will consequently influence the amplitude of the P300. Preliminary study data will be presented along with a discussion of possible brain mechanisms that underlie these findings.

T2. Neural interactions during syntactic and semantic processing of compound sentences

Myong G. Yoon and John F. Connolly

Using event-related brain potential (ERP) methods, we tested whether the putative semantic processing subsystem (the neural provenance of the N400) and the putative syntactic processing subsystem (that of the P600) function independently or they interact during the neural processing of semantically anomalous and/or syntactically incorrect compound sentences. We found that the two neural subsystems interfered each other during on-line processing of a simultaneous semantic anomaly and inflectional rule violation. On the other hand, the semantically incongruous state of a verb enhanced the neural responses (P600) to the inflectional disagreement in the same verb. The latter result shows an example of the positive mutual interaction (i.e., enhancement) between the putative semantic and syntactic subsystems.

T3. Strain and Sex Differences in MHC-Congenic Mice in Learning the Hebb-Williams Maze

L. Stanford & R. E. Brown

Using a mouse version of the Hebb-Williams Maze (HWM), we examined learning in male and female mice of two congenic strains (C57Bl/6J & B6-H-2K) for cereal reward. During acquisition males required fewer sessions than females to learn the task. There were significant differences in the number of errors over the 12 test mazes, a maze pattern by trial interaction, sex by trial, and sex by trial by maze pattern interactions, but no strain differences. The maze was then filled with water and the mice were tested under escape motivation to reach a raised platform in the goal box. In the swim-escape test the B6-H-2K mice reached criterion earlier than the C57Bl/6J mice. In the test trials, there were significant main effects for maze pattern and trial as well as the interaction of maze pattern by trial, but no strain or sex differences. These data illustrate the strength of the HWM to detect differences in learning and memory.

T4. Antisense oligonucleotides decreases the phase shifting ability of light

Renee Duquesnay¹, Harry A. Robertson¹ and Benjamin Rusak^{1,2} Department of Pharmacology¹ and Psychology²

Circadian rhythms are synchronized primarily by cues from daily light-dark cycles. Light that occurs at a time of day when it can shift (and therefore synchronize) the internal circadian clock also causes increases in gene expression in cells of the suprachiasmatic nucleus (SCN). We have investigated whether blockade of this increased gene expression locally in the SCN can alter the phase-shifting responses of the circadian system of hamsters to light. Hamsters were implanted with indwelling cannulas aimed at the SCN and were housed in constant darkness. Hamsters received either control injections or injections of antisense oligonucleotides for the IEGs c-Fos and JunB into the SCN at circadian time (CT) 12 and were exposed to a 30 min light pulse at CT 18.

Preliminary results indicate that the antisense oligonucleotides can reduce the phase-shifting effects of light, and suggest that gene expression induced by light plays a role in mediating light effects on the circadian pacemaker.

T5. Whose turn is it? Using impoverished social environments to explore human interaction.

Jeff Hancock

Contemporary analyses of human language use portray conversations as a hierarchically organized process requiring coordination of linguistic, paralinguistic and semantic information at several different levels of activity (e.g. Clark, 1996). One implication of this hierarchical model of human communication is that activities identified at lower levels of coordination (e.g. establishing predictable temporal and sequential patterns of signals, manipulating joint attentional states) must be implemented in order for higher level activities to emerge (e.g. agreed upon knowledge states and social attributions). Fortunately, in most face-to-face interactions mature speakers display exquisite control over the lower level activities in this hierarchical collaborative process. However, more impoverished social contexts (e.g. immature speakers, computer-mediated communication) may undermine the effective coordination of the lower level activities with the implication that emergent mutually agreed upon knowledge states and social attributional processes will also be disrupted. In this presentation I will describe my initial attempts to manipulate lower level communicative activities within the context of computer-mediated, text-based interactions.

T6. The Voice from Within: Parasitic Manipulation of Host Hormones and Behaviour

Kelly Shoemaker and Shelley Adamo

When the larvae of the parasitic wasp, *Cotesia congregata*, emerge from their parasitized host, *Manduca sexta*, the host exhibits an irreversible decline in feeding and locomotion. The decline is not due to debilitation. In a search for the mechanism(s) underlying the lack of sensory responsiveness, we are measuring the levels of biogenic amines, such as octopamine, in the central nervous system of both control and parasitized *M. sexta* using HPLC-ED (high performance liquid chromatography with electrochemical detection). Octopamine is thought to regulate behavioural responsiveness in insects. In our experiments, we allow *C. congregata* to parasitize *M. sexta* on the first day of the fourth instar. Then, at various times during development, we dissect out the ventral nerve cord of control and parasitized *M. sexta* and determine levels of biogenic amines. Preliminary data suggest that octopamine decreases and dopamine increases in concentration during parasitism.

T7. How Pain Catastrophizers Cope with Stressful Life Events

Nadine H. F. Rossy and M. J. L. Sullivan

Individuals differ markedly in the degree of distress they experience when faced with an aversive situation. Research has shown that individuals who 'catastrophize' are more likely to experience heightened pain and emotional distress than individuals who do not catastrophize. While the relation between catastrophizing and pain has been studied extensively, little is known about how pain catastrophizers cope with non-painful stressful situations.

The present study examined the relation between catastrophizing and methods of coping with stressful major life events. Approximately 150 undergraduates were asked to fill out the Ways of Coping questionnaire, the Pain Catastrophizing Scale and the Pain Self-Efficacy Scale. Following completion of these measures, participants were asked to immerse one arm in ice water for one minute. Consistent with previous research, we predicted that Catastrophizing would be associated with heightened pain intensity during the ice water immersion. Based on research showing a relation between catastrophizing and depression, we predicted that catastrophizers would report greater use of avoidant and escape strategies to deal with non-painful life stressors. No predictions were made about the relation between the subscales of the Ways of Coping Scale and responses to the ice water immersion.

T8. The Prevalence of Cognitive Impairment and Dementia as defined by Neuropsychological Test Performance

Fernando A. Larrea & John D. Fisk

The Canadian Study of Health and Aging (CSHA) estimated the prevalence of dementia for those aged 65 and older to be 8%. Other studies have produced both higher and lower prevalence estimates. Methodological factors that may contribute to these differences include: the use of, or the reliance on, neuropsychological testing, the consideration of functional impairment as a criterion of dementia, and the inclusion of the category of cognitively impaired, not demented (CIND) in the diagnostic classification. In order to examine these issues we conducted a reanalysis of the CSHA database for those individuals which completed neuropsychological testing. If the diagnosis of dementia required only the presence of impairments on neuropsychological testing, an increased prevalence estimate of dementia relative to the final (i.e. consensus) diagnostic opinion resulted. Adding functional impairment for the diagnosis of dementia reclassified many of the individuals as CIND and reduced this discrepancy in the prevalence estimates of dementia. However, the prevalence of CIND as defined by neuropsychological testing was higher than that defined by consensus diagnosis. The implications of these findings for neuroepidemiological studies of dementia and CIND will be discussed.

T9. Vancouver Unimanual Tactile Word Identification Test (VUTWIT) ability in patients with schizophrenia: relationship to olfactory function.

Good KP, Martzke JS, Lopeter W, Honer WG, Kopala LC.

Previous research has shown that olfactory identification deficits are common in patients with schizophrenia. Whether the olfactory deficits are specific to the olfactory system or represent more generalized processing abnormalities has not yet been resolved. Task complexity, generalized memory dysfunction and attentional compromise appear to be unrelated to olfactory deficits. However, comparison tasks used in these studies differed from the olfactory tasks in their reliability. The purpose of this study was to investigate the specificity of olfactory identification deficits in patients with schizophrenia by utilizing two analogous tasks of identification ability which were psychometrically similar, but differed in the sensory modality tested.

The University of Pennsylvania Smell Identification Test (UPSIT) was administered unimanually and the Vancouver Unimanual Tactile Word Identification Test (VUTWIT) unimanually to 24 patients with schizophrenia and 44 normal controls. Controls outperformed patients on right and left UPSIT and VUTWIT. Significant Spearman correlations were observed between left VUTWIT and right and left UPSIT. These data suggest that in patients with schizophrenia, olfactory identification deficits may not be olfactory in origin, but rather may reflect other processing difficulties.

T10. VRML Analyses of Movement

Ben Tremblay

The complexity of movement demands representations that allow the investigator to dissect individual movement properties, and also establish rules for how these properties combine in space and time. We shall show preliminary representations of movement made possible by computer VRML methodologies. Eshkol-Wachman choreography systems of individual movement parameters can be modeled, and thus analyzed.

T11. The application of event-related potentials (ERPs) in the assessment of receptive language functions

Ryan C.N. D'Arcy, John F. Connolly, Gail A. Eskes, Stephen J. Phillips, & Edmund R. Harrison

While there is a long and distinguished history of basic research with event-related potentials (ERPs), more concentration on the clinical applications of this technique is paramount. Language is well suited for investigations of an applied nature because various ERP components associated with linguistic processing have been identified. Accordingly, we have explored the feasibility of adapting standardized neuropsychological measures of language for computer presentation with ERPs. These investigations have demonstrated that ERP components related to speech perception and recognition memory can be elicited by standardized test stimuli and utilized to evaluate language independent of behavior. Specifically, studies of vocabulary and comprehension have demonstrated that: 1) test items within an individual's ability can be differentiated on the basis of their neural responses; and 2) these experimental findings have been demonstrated at the individual level, thus maintaining clinical applicability. The potential implications of this novel technique will be discussed along with the presentation of recent experimental findings from the Psycholinguistic Assessments of Language Processing in Aphasia (Kay et al., 1992).

T12. Gaze following and the control of attention in infancy

Chris Moore

In this talk I will consider the development of gaze following during infancy. I will suggest that the phenomena of gaze following are best interpreted in terms of developing attentional control. The infant's own attention is initially controlled exogenously by a particular set of social stimuli but gradually comes under more endogenous control. Only towards the end of infancy, does such endogenous control provide us with a warrant to claim that infants understand the attentional states of others.

T13. Stress and Finickiness

Tara L. Fidler and Vincent M. LoLordo

Finickiness, suppressed consumption of a novel bitter solution relative to baseline consumption levels, has been observed following stress in humans and animals. In studies reported previously (for example see Job and Barnes, 1995, *Behavioral Neuroscience*, 109, 106-116) the finickiness effect has been reported only following the sorts of shock parameters that result in learned helplessness, 100+ shocks of 2-60 s in duration. Our rats demonstrated finickiness for novel quinine 40 minutes after a shock session, consisting of 60 shocks of 2-s duration on a FT-60 s schedule. The effect was reinstated 27-28 days later with only 10 shocks. Uncontrollable shock stress may increase the aversiveness of bitter tastes.

T14. Signaling Benevolence: An Evolutionary Perspective on the Encoding and Decoding of Altruism

William M. Brown, Boris Palameta & Chris Moore

Emotions associated with benevolence can be viewed as evolved commitment devices that help solve social dilemmas which require trust (Frank, 1988). Non-kin altruism may be evolutionarily stable if these emotions are signaled and decoded accurately. It was hypothesized that altruists produce involuntary facial signals which are detectable by others. One-minute video-clips were constructed of 5 self-reported altruists and 5 self-reported non-altruists. Altruist/non-altruist pairs were videotaped while playing the game "Mastermind" jointly against the experimenter. In one condition the altruist received credit for game performance; while in the other condition the non-altruist received credit. Under blind conditions, one-hundred and sixty eight undergraduates rated the 10 video-targets' helpfulness, concern, attentiveness and expressiveness. Results suggest that individuals can detect altruists. Altruists were viewed as significantly more attentive and concerned than non-altruists when playing the game for their partner's benefit. Non-altruists were rated significantly more attentive and concerned than altruists when playing the game for their own benefit. Regardless of condition altruists were viewed as more helpful than non-altruists. Results will be discussed in light of current conceptual models explaining the evolution of altruism among non-relatives.

T15. An Attention Based Negative Priming Effect : Really!!

John Christie

Negative Priming (NP) was once thought to be the way to prove a hypothesis first put forth by Treisman (1964). If one ignored an item, while attending another, the ignored item would be inhibited. For the first time hundreds of researchers started to study inhibitory effects of attention through the use of NP. However, since NP was discovered nearly all of the founders of the phenomenon have abandoned this attentional explanation in favour of a memorial one. Those who have not abandoned the inhibitory attentional explanation have decided to stick with models that can only account for a fraction of the available data. While current memorial explanations are difficult, if not impossible, to defeat, one NP effect has been found that is best explained by attentional orienting. The reasons for the generation of this particular effect may also allow one to design other experiments to examine inhibition.

T16. Development of a brief battery of attentional tasks.

J.R. Blackburn, R.J. McInerney, & R.M. Klein.

In order to determine whether attentional capacity changes as individuals spend time playing a video lottery terminal (VLT), we have developed a battery of computerized attentional tasks that may be administered both before and after a session of VLT play. Six previously characterized tasks have been modified to make them appear similar to each other and to make them more like games for our gambling subjects. The six tasks were a continuous performance task (CPT), a CPT-AX task, a Simon effect task, a flanker task, a Rapid Serial Visual Perception (RSVP) task, and a stop signal task. A factor analysis of data from twenty undergraduate subjects will be presented, indicating whether there are identifiable factors of attention and whether the battery can be shortened by removing redundant tests.

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T17. Post-Polio Syndrome: A developmental disorder resulting from childhood poliomyelitis infection which has implications for neuroscience and psychology.

Richard Brown

Millions of people who contracted the poliomyelitis virus as children survived this disease and grew into adulthood without further polio symptoms. But as many of these former polio victims turn 40 or 50, they suffer muscular weakness caused by the deterioration of their motor neurons. Post-polio syndrome was first named as a disease in 1983 and is the most common new neuro-motor disease in the world. There are three theories about the development of post-polio syndrome: (1) it is due to neuromotor degeneration; (2) it is an immune dysfunction; or (3) it is caused by persistence of the poliomyelitis virus. Since Sabin invented the polio vaccine in the 1950's, polio is almost unknown in developed countries and the World Health Organization is attempting to vaccinate everyone in the world before 2000 so that this disease is eliminated. Unfortunately, new cases of polio are emerging and those who survived polio as children are being victimized a second time with post-polio syndrome. Many doctors have not been trained to recognize polio or post-polio syndrome. This talk examines some of the challenges for neuroscience and psychology in understanding the behavioural and neuro-muscular changes presented by post-polio syndrome.

T18. Melatonin and Light Effects on Circadian Rhythms and Thermoregulation

Xiaowei Song and Ben Rusak

Internally generated biological rhythms are synchronized to the external day principally by light-dark cycles. Melatonin is secreted by the pineal gland at night and it too can influence rhythm synchronization, apparently by having effects opposite to those of light. Melatonin has, therefore, been used to treat disorders of rhythm entrainment in humans. The normal nocturnal decline in body temperature in human appears to be related to melatonin secretion. Light exposure at night reverses the nocturnal temperature decline by inhibiting melatonin secretion, an effect that can be opposed by giving exogenous melatonin. Melatonin administration also appears to attenuate the phase-shifting effects of nocturnal light. Since it is not clear how melatonin and light interact to affect circadian rhythm synchronization and thermoregulation, we are studying light and melatonin interactions using rodent model systems, in which levels of the two factors can be manipulated for both nocturnal and diurnal species. Changes of core body temperature, hypothalamic temperature, and locomotor activity in response to light exposure and melatonin administration are being monitored continuously. Preliminary findings related to the effects of light and melatonin on body temperature, activity, and circadian rhythms will be discussed.

Importance of Melatonin in Circadian Rhythms

T19. The influence of parafoveal identification on letter detection accuracy revisited

Jean Saint-Aubin and Raymond M. Klein

When asked to detect target letters while reading a text, subjects miss more letters in highly frequent function words than in less frequent content words. According to the parafoveal-processing hypothesis, omissions occur when a word is identified parafoveally, instead of being fixated. Function words would be more likely to be identified parafoveally, because their high frequency allows identification on the basis of minimal visual information. This hypothesis has been tested with rapid serial visual presentation in which fixation was encouraged by presenting words one at a time on a computer screen. Detection accuracy was compared for target letter t embodied in the or in content words. Eighteen subjects read three prose passages, each presented at a different speed : 200, 350 or 500 msec per word. Contrary to the parafoveal-processing hypothesis, results showed more omissions for the at the fastest presentation rate. However, due to ceiling effects, there were no effects at the slower presentation rates.

T20. Relationships of Anxiety Sensitivity and Dimensions of Alexithymia in a Young Adult Sample

Sherry H. Stewart, Heather Devine, and Margo C. Watt

The relationships between anxiety sensitivity (AS) and alexithymia dimensions were examined in a non-clinical sample of young adults, after controlling for item redundancy between the Anxiety Sensitivity Index (ASI) and the 20-item Toronto Alexithymia Scale (TAS-20). A sample of 238 undergraduate students completed the ASI and the TAS-20, as well as measures of trait anxiety and panic attack history. As hypothesized, high AS students ($n = 36$) scored significantly higher than low AS students ($n = 41$) on the two TAS-20 subscales suspected of sharing a functional relationship with AS (i.e., difficulty identifying emotions; difficulty describing emotions), whereas no AS group differences were found for the TAS-20 subscale believed to be functionally unrelated to AS (i.e., external-oriented thinking). Moreover, the pattern of relationships between ASI scores and alexithymia dimensions persisted following the removal of TAS-20 items identified as potentially conceptually redundant with the ASI, confirming the prediction that the relationship between AS and alexithymia is not merely an artifact of item redundancy. The results also revealed that individuals who both experience frequent anxiety (trait anxiety) and who greatly fear their anxiety symptoms (AS) appear most likely to use alexithymic coping strategies. Finally, the results showed that high AS in the absence of a history of panic attacks appears sufficient to motivate the use of alexithymic coping strategies. Potential clinical implications of the findings will be discussed.

T21. Waist-to-Hip Ratio and Perceptions of Female Attractiveness

Boris Palameta, Christina Staples, & Debbie Bird, Psychology Department, St. Thomas University, Fredericton, NB

In a series of influential papers, Dev Singh has argued that human males are selected to use female waist-to-hip ratio (WHR) as a criterion for mate choice, because low WHR is an honest signal of health and fertility. This paper reports the results of experiments designed to test some of Singh's hypotheses. A computer program (Adobe Photoshop) was used to manipulate the WHRs of photographed figures in 4 ways: 1) remove weight from waist (reduce WHR); 2) remove weight from hips (increase WHR); 3) add weight to waist (increase WHR); 4) add weight to hips (reduce WHR). Consistent with Singh's theory, underweight and average-weight figures were rated as more attractive when weight was removed from the waist rather than the hips. However, the reverse was true for overweight figures. Furthermore, despite their low WHRs, figures with experimentally enlarged hips were not perceived to be more attractive than figures with enlarged waists. Gender differences in attractiveness ratings were also investigated.

T22. "Yet a little sleep, a little slumber, a little folding of the hands to sleep" (Proverbs 6:10)

B. Rusak, J. LeBlanc, R. Morehouse, M. Rajda, S. Thompson, S. Kutcher

In the clinical psychiatric literature (including the DSM IV), a sleep disorder known as Delayed Sleep Phase Syndrome (DSPS) has been described. It is characterized in both adolescents and adults as a disorder of the timing of sleep and waking, in which sleep onset is strongly delayed when the individual is allowed to sleep at his or her preferred time, but is structurally normal. Under usual work or school conditions, however, sleep is severely disrupted because the individual is essentially incapable of shifting from the very delayed timing to more socially acceptable patterns. It is generally regarded as a disorder of circadian system entrainment. DSPS can have very serious consequences for school attendance, school performance and job opportunities. The incidence of this disorder has been estimated in several studies, but these estimates have varied over a couple of log units. We undertook an epidemiological survey of ~1600 adolescents in Halifax Regional Municipality to assess sleep disorders, mood disorders and their relation to reported school performance, with an emphasis on identifying the incidence of DSPS in this population. While most of the data remain to be analysed, we will report on our preliminary findings and discuss the difficulties of definition and diagnosis in this area.

T23. Exploring The Potential Link Between Saltation, Reading, And Perceptual Grouping

David I. Shore and Raymond M. Klein

We explored the auditory variant of the saltation illusion (the mislocalization of a brief stimulus followed by a stimulus at a different location) and developed a unique measurement tool. Prior research has shown that auditory saltation is more robust for adult dyslexics and, as such, has been proposed as a measure of temporal processing. Subjects participated in three tasks: a subjective rating of the saltation illusion, a click enumeration task, and the Olson reading tests. Click enumeration is proposed to provide an index of perceptual grouping while the Olson test provides separate measures of orthographic and phonologic coding. Individual differences in these three measures were compared. The complex interactions between these three measures are discussed in relation to proposed links between temporal processing, perceptual grouping and reading.

T24. The joy of "hide-and-seek": Why is it harder to find people than objects within a false belief task

Doug Symons

In order to better understand the interface between social and cognitive development in the preschool period, current research in my lab has focused on false-beliefs held for social referents. The paradigm we have developed is based on the standard Maxi task in which a participant is asked where a story-character will look for a caregiver or object relocated within the task without the character's knowledge. We are interested in why children do more poorly on tasks involving people compared to objects, and have varied the reason for relocation of the sought item to determine why. A series of studies will be presented to show that animacy and intentionality confound performance and delay standard age-effects when the sought item is a person. One explanation is that developing a theory of mind actually reduces performance when the sought-item has a mind, and ultimately, such awareness fits in with relationship constructs.

T25. A Paradigm for Investigating Olfactory Conditioning and Differential Immediate Early Gene Expression in Mice

Catherine Forestell and Heather Schellinck

The purpose of the present experiment was to determine whether mice can acquire a conditioned response to an odour paired with a caloric reinforcer. Food restricted mice were exposed to a differential conditioning procedure in which experimental animals received odour A paired with sucrose in a dish covered with bedding and odour B presented alone in a dish with bedding. Control animals were either exposed to odour A and odour B alone, or to no odour with bedding. On day 5, animals were placed in a chamber divided into three equal sections with odour A and odour B on each end of the chamber. Experimental animals spent significantly more time digging in the dish which contained Odour A than Odour B relative to control animals. On day 6, all animals received exposure to odour A, odour B or no odour, 2 hours prior to perfusion. Evidence for immediate early genes expression for c-fos was assessed in brain structures such as the olfactory bulbs, amygdala and the basal ganglia. Immunohistochemical results will be discussed.

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T26. Contralateral Dopaminergic Neurons are Involved in Generating the Reinforcing Effect of Brain Stimulation

Seth Bloom and Shinshu Nakajima

It is well known that the mesolimbic dopaminergic path plays an important role in generating a reinforcing effect of brain stimulation. The majority of studies in this field, however, dealt with only one side of the brain—the side where the electrode is implanted, and the contralateral effect has been ignored. The functional relationship of contralateral dopaminergic pathways involved in brain stimulation reward was examined. Rats were implanted with two cannulae into the VTA and the NAc contralateral to a lateral hypothalamic stimulating electrode and trained to bar-press. The frequency of stimulation pulses was systematically changed, and a frequency-response curve was plotted for each rat before and after 6-OHDA injections into the VTA. Haloperidol was also injected into the NAc before and after placing the lesion to further examine the role that contralateral dopaminergic pathways play in self-stimulation. The reward threshold increased after either a 6-OHDA lesion of the VTA or a blockade of dopaminergic terminals in the NAc. The results suggest that a contralateral pathway is involved in the reinforcing effect of hypothalamic stimulation.

T27. Depression and Substance Use Disorder Comorbidity Patterns Among Adolescents

Nancy Comeau and Darcy Santor

Objective: Since the introduction of DSM-III-R, clinicians and investigators have shown increasing interest in the study of conditions comorbid with affective disorders. Understanding depression and substance use disorder comorbidity patterns among adolescents is needed to guide treatment, research, and prevention/early intervention approaches. Method: 1) The literature on the comorbidity of depression and substance use disorder among adolescents from the past 10 years was reviewed to explore the most prevalent patterns of depression-substance use disorder comorbidity; 2) examine the correlates and longitudinal predictors of comorbidity; and 3) determine the extent to which comorbid patterns convey unique information concerning etiology and treatment. Results: a body of literature concerning the comorbidity of depression and substance use disorder, with a substantially smaller amount of data concerning other comorbid conditions. In many areas the literature was sparse, and pertinent questions concerning comorbidity patterns remain unexplored. Nonetheless, available data warrant that interventions among adolescents dealing with these comorbid conditions are likely to be quite important to society. Conclusions: Additional studies of the frequency of comorbidity and associated factors are greatly needed, to include studies of differential effects of treatment of adolescents with various comorbid depression and substance use disorders, as well as of depression and substance use disorder adolescents who differ in etiological factors.

The concept of risk - risk for substance use
- risk of substance use
The connection to drug dependency.

**Department of Psychology and Neuroscience
24th Annual In-House Psychology Convention**

Keynote Address

Dr. John Fentress

**WOLF WALTZES, MOUSE MUSIC AND BRAIN BOXES:
CAN THERE BE A MEETING OF THE MINDS?**

Animal actions, as well as specifically human actions, come in many different guises. It's the flow of actions in time that gives behavior its essence, and thus its basic challenges for analysis. How do we thus isolate properties of action in time, so that we can also capture the dynamic linkages among these individual properties? My approach is broadly ethological. I am interested in how different species do what is special to them, and then how we might seek rules to generalize across this diversity. Subsequent questions are how these actions emerge in development, and how they relate to operations of brain. I have been privileged to work with some fascinating creatures, and to struggle with the problem of conceptualizing as well as analyzing the problems they offer. To do this I have often reached to areas outside of traditional science; thus waltzing wolves and musical mice. Where do these reaches lead us? often they lead up tangled garden paths. Occasionally they open up new lines of thought, and its these lines of thought I wish to emphasize here. My current favorite line of thought is that the dynamics of action involve a poorly understood (and necessarily dynamic) balance between autonomy and interdependent control operations. My animal (and occasionally human) friends have lead me into the brain. How does the brain both separate and combine events in time? Tidbits of information on basal ganglia and cerebellum might provide some clues. I conclude with a simple statement of appreciation to this department. It is rewarding to be associated with faculty, students and staff who support personal inquiry within an atmosphere that has also been supportive of non academic aspects of my stay at Dalhousie.

Thursday, April 30th/1998
3:30 p.m.
Life Sciences Centre Room 240
Dalhousie University

T28. Development and evolution of Ascidian nervous systems

Alison Cole

I am interested in mechanisms of neuronal differentiation in development. Ascidians are closely related to vertebrates and are thought to be the closest extant form resembling the early chordate common ancestor. The combination of invertebrate characters with their vertebrate heritage make ascidians ideal for studies in neuronal development. I hope to extend a previous study tracing the cell lineage development in larval *Ciona intestinalis* through confocal reconstructions with various immuno-probes, as well as further identifying cells with regard to their transmitter phenotype. This work will establish whether or not lineage and cell type differentiation are related. I also hope to explore the evolutionary relations between the ascidian groups by comparing the development of the dorsal nervous system across different ascidian species.

T29. On the causes, effects, functions and neural implementation of "inhibition of return"

Ray Klein & Tracy Taylor

Immediately following an uninformative peripheral stimulus (cue) there is a short-lasting facilitation of response times in the region of the cue which is followed by a longer lasting inhibition. This inhibitory effect of the cue upon subsequent responses to targets was named inhibition of return (IOR) to capture the dominant interpretation that, after having oriented to a location in space, attention is inhibited from returning there. This phenomenon has been the subject of intense interest because of its putative role in improving our ability to forage the visual environment for consequential information (Posner & Cohen, 1984; Klein, 1988; see Taylor & Klein, in press, for a review). I will present evidence from various methods in cognitive neuroscience which converge on: oculomotor programming as the cause of the inhibition, localization dependent processes as affected by the inhibition, enhanced search efficiency as one function of the inhibition, and the superior colliculus as the main neural structure involved in generating the inhibition.

T30. Effect of Male Removal on Gestation Length in Female California Mice (*Peromyscus Californicus*)

A.W. Lee, D. Cantoni & R.E. Brown

California mice are a monogamous species that show biparental care. In mated pairs, the presence of the male facilitates reproductive success by increasing pup survival and reducing the female's gestation length. Under wheel running conditions for food, females living with their mate have a mean gestation length of 33 days. Females which have their mates removed 2 days postpartum, have a mean gestation length of 52.5 days, suggesting that these females had delayed implantation.

The present study investigated the effect of male removal near the middle (Day 17) or end (Day 32) of the gestation period. Females living with their mate with ad lib food, have a mean gestation length of 34 days. Females which have their mates removed on Day 32 of gestation have a mean gestation length of 36 days. Females which have their mates removed in on Day 17 of gestation either have a mean gestation length of 36 days or fail to give birth. These results suggest that the presence of the male is important throughout the gestation period in maintaining a normal, gestation length in female California mice. Aspects of the male's behavior during parturition which may facilitate parturition will be discussed.

T31. Copy Cat: Spatial Localization Deficits in Cats with Experimentally Induced Amblyopia are Identical to Those Experienced by Human Amblyopes

Guy Gingras and Donald E. Mitchell, Department of Psychology, Dalhousie University

Although human amblyopia is defined as uncorrectable poor visual acuity, not all forms of acuity are affected equally. The worst deficits are observed on tasks that measure the ability to locate differences in the position of objects in space. This research had as its goal a comparison of the performance of normal cats and humans on the same localization task (a test of the ability to detect a misalignment between 3 Gaussian blobs) and to document the deficits on this task of cats with various forms of early visual deprivation (monocular deprivation or strabismus). Deprived cats demonstrated localization deficits that changed with spatial scale (blob size) in a manner identical to that observed in human amblyopes. Moreover, like human amblyopes, the spatial deficits on the alignment task were considerably larger than the deficits of grating acuity. The results justify the use of these animals models of human amblyopia.

T32. Instant identification of popular songs: Insights into auditory memory representations

E. Glenn Schellenberg¹, Paul Iverson², Margaret C. McKinnon³ ¹Dalhousie University, ²University of Washington, ³University of Toronto

We examined memory representations for popular songs by testing listeners' ability to identify brief excerpts from familiar recordings. Although songs are defined solely by pitch and temporal relations, such representations might also contain information about timbre (sound quality). Listeners were required to match 200- or 100-ms excerpts with the song titles and artists. Performance was well above chance levels for 200-ms excerpts, and poorer but still better than chance for 100-ms excerpts. Performance fell to chance levels when dynamic information was disrupted by playing the 100-ms excerpts backwards, and when high-frequency information was omitted from the 100-ms excerpts; performance was unaffected by the removal of low-frequency information. In sum, successful identification required the presence of dynamic, high-frequency spectral information. This finding indicates that the timbre of specific recordings is encoded into memory representations of popular songs.

T33. Space, the Final Frontier (for Gap Detection)

D.P. Phillips, S.E. Hall, I.A. Harrington and T.L. Taylor

The detection of brief periods of quiet ("gaps") in the ongoing stream of speech is extraordinarily important for the identification of the stop consonants. In a previous lifetime, we argued that the perceptual task of detecting such a gap can exploit either a discontinuity detection mechanism, or a relative timing mechanism, depending on whether the noises that bound the gap are the same or different. That is, the gap detection operation can either be executed within a particular perceptual channel, or as a cross-correlation of the activity in different channels. In our previous work, we forced the use of different perceptual channels by using gap markers that stimulated different populations of peripheral neurons. This year, we describe the gap detection behavior of listeners in an experiment in which the gaps were bounded by markers that stimulated the same sets of peripheral neurons, but different central perceptual channels.

T34. Gaps: The Next Generation

S.E. Boehnke, D.P. Phillips, and S.E. Hall

In the previous abstract, Phillips et al. showed that separating the leading and trailing noises of a gap stimulus in space results in the gap detection task becoming a "between-channel" one. We thus suspected that the gap detection paradigm might be used to determine the tuning of spatial channels in human hearing. Consequently, I have looked at the effect of systematic changes in the spatial location of the leading and trailing markers of a gap stimulus. This task requires a relative timing operation to occur between two populations of central neurons coding for spatial locations of the leading and trailing markers. Intriguing results point to the existence of three spatial channels - left hemifield, right hemifield, and midline. Equally fascinating however, is that these results map almost perfectly onto the underlying neurophysiology of the central auditory coding of the binaural cues for sound location.

T35. Presynaptic adrenergic receptors inhibit REM sleep by attenuating pontine cholinergic transmission.

Kazue Semba, Department of Anatomy and Neurobiology

Mesopontine tegmental cholinergic neurons are thought to be responsible for inducing REM sleep by releasing acetylcholine (ACh) in the pontine reticular formation (PRF). However, many of these neurons appear to be active (thus, potentially releasing ACh in the PRF) not only during REM sleep but also during wakefulness. One explanation for this apparent paradox is that, during wakefulness, ACh release in the PRF is strongly inhibited by noradrenaline acting on presynaptic adrenergic receptors on cholinergic terminals. Noradrenaline is likely to be released in the PRF during wakefulness by locus coeruleus neurons. Our data using behavioural, in vitro electrophysiological, and in vivo microdialysis techniques support this hypothesis, and suggest that presynaptic mechanisms might play an important role in the control of behavioural state.

T36. The Embryonic Brain of *Drosophila melanogaster*, or A Portrait of the Fruitfly as a Young Egg

Jason Emsley

The fruitfly, *Drosophila melanogaster*, has long been used a vital tool for neuroscientific research of a genetic and anatomical nature. Because of its stereotypical pattern of development and its relatively simple nervous system, *Drosophila* is a powerful tool for understanding the relationship between gene and neurotransmitter expression and development. I will present several views of the development of this simple nervous system, considering how information about gene and neurotransmitter expression can be related to the actual anatomy of the developing CNS. Semi-thin sections of embryonic tissue have been serially aligned and, from these sections, I have developed a computerised, three-dimensional anatomical atlas of the *Drosophila* nervous system. Images of this atlas will be considered, alongside images produced via fluorescence immunohistochemistry and confocal microscopy.

T37. Behavioural Differences in MHC-Congenic Mice: The Step-Down Passive Avoidance Test

Tim Bredy, Will Moger and Richard Brown

Although there are a multitude of physiological and pathological traits in which the major histocompatibility complex (MHC) might influence behaviour (Ivanyi, 1978, *Proc. R. Soc. Lond. B*, 202, 117-158), MHC-congenic mice have primarily been tested for mate selection and individual recognition by odour (Yamazaki et al., 1976, *J. Exp. Med.*, 144, 1324-34; Manning et al., 1992, *Nature*, 360, 581-583). The present experiment investigated the possibility that two strains of MHC congenic mice (C57Bl/6J vs. B6 H-2K) would differ in their ability to learn and remember a passive avoidance task. Glucocorticoids influence learning and memory (Smith, 1996, *Beh. Brain Res.*, 78, 25-36) and there are strain differences in hippocampal glucocorticoid binding capacity in mice (Patacchioli et al., 1990, *Beh. Brain Res.*, 37, 185-188). Plasma corticosterone concentrations were therefore measured using radioimmunoassay following the passive avoidance test. There was a significant strain difference in the ability of the no shock control group to habituate to the step-down passive avoidance apparatus that can be attributed to a strain difference in learning. The results indicated no strain differences in retention. Corticosterone levels are presently being analyzed. Strain differences in passive avoidance learning and in corticosterone levels may explain some of the MHC-related strain differences in mate selection, intermale aggression and other behaviours.

T38. The tenacious blink

E. McLaughlin, R. Klein & D. Shore

When stimuli (e.g., letters) are presented in rapid succession, individuals are typically poor at identifying a target (probe) that follows an earlier target (prime) by 100-500 ms. Four studies investigated the nature of this perceptual failure, which is known as the "attentional blink". These studies show that the strength and quality of the attentional blink is very robust. Studies 1 and 2 compared two distinct methods of measuring the blink: letters presented within a stream of numbers versus single letters followed by pattern masks, with the ISI between letters held constant across studies. As predicted, blink strength was highly correlated within subjects across studies. Studies 3 and 4 investigated whether manipulating the quality of the physical evidence for the target stimuli (either prime or probe) would affect the attentional blink. These manipulations had the expected effect upon the processing of the degraded target (accuracy was a monotonic function of the quality of the evidence), but contrary to our expectations, manipulating the difficulty of extracting the identity of either the prime or probe target had no impact on the strength of the blink. The implications of these findings for understanding the mental processes underlying the blink will be discussed.

T39. An Examination of Differences in the Time-Course of Oxazepam's Effects on Implicit Versus Explicit Memory

Susan Jerrott

The present study was designed to examine the effects of oxazepam on implicit versus explicit memory processes, as a function of this drug's time course. The effects of oxazepam (30 mg) or placebo on directly comparable tests of implicit memory (word-stem completion) and explicit memory (cued-recall) were examined at three time points: 100 minutes post-drug (prior to the peak plasma concentration of oxazepam; i.e., "pre-peak" condition), 170 minutes post-drug (close to peak; i.e., "peak" condition) or 240 minutes post-drug (following peak; i.e., "post-peak" condition). Sixty volunteers were randomly assigned to either the drug condition or the placebo condition in a double-blind design and were tested on both memory tests at one of the three time points. In the "pre-peak" condition, oxazepam impaired cued recall performance but did not impair priming. In the "peak" condition, oxazepam impaired performance on both memory tasks. In the "post-peak" condition, cued recall performance in the oxazepam group remained significantly impaired. However, oxazepam-induced impairments in priming were only marginal, suggesting that oxazepam-induced impairments in implicit memory processes begin to wane following peak drug concentrations. These findings support the hypothesis that benzodiazepines exert time-dependent effects on implicit memory processes. The results also support the theoretical distinction between implicit and explicit memory processes, since the directly-comparable implicit and explicit tasks showed different impairment curves over time.

T40. Life stress and depressive severity in adolescents with mood disorders

Sonia H. Greaven, Darcy A. Santor & Vivek Kusumakar

This study investigated the occurrence of life stressors with mood disorders in an adolescent population. Forty-two clinically-referred adolescents with primary mood disorders (30 major depressed, 12 bipolar disordered), between the ages of 12 and 21 ($M = 16.06$ years) participated. Each participant underwent a systematic lifetime diagnostic interview (K-SADS-PL), two measures of current depressive severity, and a computerized inventory of life stressors within the past two years. Results indicated that the total number of life stressors, as well as the severity of the stress associated with these, were related to current depressive severity. Examination of correlations with specific stressor categories revealed that overall, minor hassles, as opposed to major life events, were related to depressive severity.

T41. Neuraltransplantation for Parkinson's disease and Gene therapy for malignant brain tumors

Ivar Mendez

No abstract

T42. Cognitive Sequelae of Cancer and its Treatment in Children

Veronique Mackenzie-Bourne and Richard E. Brown

We reviewed 60 papers on the cognitive and neurologic sequelae resulting from the three main cancer treatments, alone and in combination, for the treatment of brain tumours and acute lymphoblastic leukemia. The initial goal of this paper was to examine studies concerning long-term effects in cancer survivors under fifteen years of age and over the age of sixty-five. The main body of literature, however, focuses on children. Long-term cognitive effects in older survivors of childhood and adult cancers have not been studied. The research reviewed suggests that some forms of cancer treatment are worse than others for resulting cognitive sequelae, e.g. whole-brain irradiation and some chemotherapeutic agents (e.g. methotrexate), especially in children under 3 years of age. However, a direct link has yet to be found between cancer treatment and long-term cognitive dysfunction. This is due, in part, to a lack of consistency in research design, method, and populations studied. Research concerning the cognitive problems of treating cancer in the elderly and in determining the long-term effects of cancer treatments in elderly people who are survivors of earlier cancer must also be developed.

T43. Pavlovian Conditioning in the Circadian System: Fact or FANTasy?

Marleen de Groot and Benjamin Rusak

Environmental stimuli such as light can shift or entrain circadian rhythms and induce physiological changes in cells comprising the central circadian pacemaker (the suprachiasmatic nucleus: SCN), including altering the expression of immediate-early genes (IEGs). A recent study (Stewart & Amir, 1996) showed that a stimulus that does not normally have any of these effects can be made to do so after being paired repeatedly with a phase-shifting light pulse in a Pavlovian conditioning paradigm. We assessed whether such Pavlovian conditioning can occur in the circadian system. During training phases, a 20 min air disturbance (CS) was either paired or explicitly unpaired with a 60 min light pulse (US) twice daily for ten-and-a-half days. Behavioural and gene expression results show that the CS does not come to mimic the effects of light on the circadian system following pairing. The failure to find evidence of conditioning could not be explained by a lack of sensitivity to the CS used, since all subjects showed clear behavioural responses to the presentation of the CS during both the training and testing phases of the studies.

T44. Suppression of protein kinase C in the basal forebrain reduces the rewarding effect of brain stimulation.

Robin L. Patterson and Shinshu Nakajima

Activation of D1 and D2 dopamine receptors in the nucleus accumbens (NAc) is important for the rewarding effect of various reinforcers including brain stimulation. Blockade of D2, but not D3 or D4, receptors in this region with dopamine antagonists increases the threshold of reinforcement for brain stimulation. Activation of D1 receptors is necessary to activate D2 receptors. In the present study, the reward mechanism within NAc was studied by using *calphostin C*, a selective inhibitor of protein kinase C. The subjects were 42 male albino rats. Each of them was implanted with a bipolar electrode into the medial forebrain bundle and two cannulae bilaterally into NAc. They were trained to press a bar to receive electrical stimulation, and the threshold of reinforcing effect was measured before and after the injection of *calphostin* through the cannulae. Bilateral injection of *calphostin* into NAc increased the threshold of reinforcing effect in 2 hours and the effect reached the maximum in 4-6 hours. Recovery was observed in 24 to 48 hours after injection. Unilateral injection, either ipsilateral or contralateral to the electrode, was not effective. The results suggest that the rewarding effect of medial fore-brain stimulation is mediated by the activity of protein kinase C in NAc neurons.

Symposium

To Hebb and Beyond

Donald Hebb was one of Canada's most famous and distinguished scientists, whose contributions were manifold and writings prolific. Hebb is widely recognized for his neo-connectionist assumption that correlated firing results in improved synaptic transmission (increase in connection weight) and his proposal of the cell assemblies. Hebb was a major advocate of psychology as a science and was a major force in directing its evolution in the 20th century. In order to examine Hebb and his work in a broader context, we will attempt to situate Hebb and his ideas within history, both in terms of the people who influenced him as well as in how his ideas shaped (or failed to shape) subsequent researchers. Some of the areas influenced by Hebb include LTP, kindling, neural networks, animal learning and memory, mental imagery, stopped retinal image, pain, aging and the role of early experience in cognitive and perceptual development. A life history of Hebb connecting significant life events to the various areas of interest integrates the diverse fields in which Hebb made a notable impact. For example, the publication of the *Organization of Behavior* (1949), which was influenced by the writings of William James and Santiago Ramon y Cajal, has exerted an ever-growing influence on those interested in mind, brain and how brains implement mind.

Posters

What does that robot want from me now? Young children's abilities to repair contingent queries.

Suzanne Ferrier

The ability of two-year-olds to respond differentially to specific and general contingent queries was examined with the use of an immobile robot as the conversation partner. Twenty-one 2;9-year-old and 21 2;3-year-old children were queried (5 specific and 5 general) immediately following 10 of their spontaneous utterances in a free play session. The type of response strategy (repetition or specification) was coded for each queried utterance. Regardless of age, children tended to respond to a general query with an exact repetition of their original utterance. Age differences were found, however, in their replies to a specific query. The 2;9-year-olds most often replied with the specific information requested, whereas the 2;3-year-olds often repeated their original utterance, providing their partner with too much information. These findings suggest that it is not until 2;9 years of age that children efficiently satisfy their communicative partner's informational needs when conversational exchanges involve only linguistic cues.

Temperature Selection and Use of Torpor by the Marsupial *Sminthopsis Macroura*

Xiaowei Song* and Fritz Geiser

Department of Zoology, School of Biological Sciences, University of New England, Australia *present address: Rusak Lab, Department of Psychology, Dalhousie University

Many small mammals display daily torpor to minimize energy expenditure during the rest phase when faced with unfavorable environmental conditions. However, given a choice of thermal environments, it is not clear whether these daily heterotherms select ambient temperatures based on whether these minimize metabolic rates during torpor or during normothermia. It is also not clear whether they prefer being normothermic or torpid. In this study, we investigated temperature selection, activity patterns body temperature patterns, and the use of torpor in *Sminthopsis macroura* in a thermal gradient when food was freely available or restricted. Animals employed torpor regularly immediately after cessation of activity even when excess food was provided. Nevertheless, they selected high ambient temperature near the zone of thermoneutrality during both normothermia and torpor. Reduced food supply did not affect either temperature preference during torpor. Our study shows that *S. macroura* selected a high ambient temperature to reduce energy costs during normothermia. Nevertheless, daily torpor was frequently employed at the high temperature, especially when food was restricted.

Measurement of Pain in Non-verbal, Cognitively Impaired Individuals

Lynn Breau, Patrick J. McGrath, Carol Camfield, Christina Rosmus, G. Allen Finley

A preliminary investigation of the properties of the Non-communicating Children's Pain Checklist was conducted. Caregivers of 38 individuals with Cerebral Palsy, Autism/PDD or a neurological disorder completed the checklist at home: twice when that person was in pain, once when they were in distress, and once when they were calm. Caregivers also completed the Checklist concerning their participant's past pain behaviour in general. Analyses were conducted to assess the Checklist's internal consistency, concurrent validity, sensitivity, specificity, and reliability. The results indicate that, when used by caregivers on actual events in a natural environment, the checklist can discriminate between calm and pain, is reliable over time, correlates with a visual analogue measure of pain, and is internally consistent. No differences were found between diagnostic groups, although reported behaviour did vary as a function of the relationship of the caregiver to the individual, and that individual's level of verbal comprehension.

Evidence Based Decision Making: Construct Validity of a Sodium Amobarbital Recognition Memory Protocol

McGlone, J., Black, S.E., Evans, J., Grantmyre, A., Moscovitch, M., Parkin, A., Sadler, M., Sita, A., Squires, E., Stuss, D., and Wilson, B.A.

Purpose: We tested whether the Intracarotid Amobarbital Protocol had criterion and concurrent validity. It was hypothesized that a recognition memory test designed for intracarotid injections and used to predict the risk of global amnesia prior to an elective temporal lobectomy should identify persons who are severely amnesic due to other neurological causes. Speech tasks would be unaffected by amnesia, but would identify aphasia. The impact of aphasia on memory outcome would be negligible.

Methods: Fifteen persons with severe amnesia and one aphasic were administered a four-item yes/no recognition memory protocol and a speech protocol. No drug injection occurred. Standardized neuropsychological tests were used to divide the amnesic group into those with Global Amnesia (i.e., retain no ongoing memories), Dissociated Amnesia, and Amnesia plus other neuropsychological deficits (i.e., Non-Dissociated).

Results: Two Global Amnesic persons obtained scores at or below chance (i.e., failed) on the memory protocol. Unexpectedly, the remaining 13 severely amnesic persons (Dissociated and Non-dissociated), obtained passing memory scores. Amnesia had no impact on the speech protocol. An aphasic individual passed the memory protocol and was impaired on the speech protocol.

Conclusions: This IAP memory protocol showed good criterion validity in identifying Global Amnesia, but it was insensitive to other disabling, severe amnesic disorders. Discriminant validity was shown by amnesic persons performing well on IAP speech tasks. These results suggest that this IAP protocol might have validity in predicting a post surgical Global Amnesic disorder, but it did not identify and therefore could not predict other severe amnesic disorders

The Effects of Alcohol and Alcohol Expectancy on Responses to Hyperventilation in High Anxiety Sensitive Individuals

Alan MacDonald, Rochelle Hutson and Sherry Stewart

High anxiety sensitivity (AS) levels are significantly associated with increased beliefs about the tension-reduction properties of alcohol (Karp, 1993). The present study examined the effect that these alcohol expectancies might have on alcohol's reactivity dampening effects in high AS young adults (Baker, MacDonald, Stewart & Skinner, under review), in addition to alcohol's pharmacological anxiolytic effects. A three cell beverage condition design was used to assess the contributions of expectancy vs. expectancy plus pharmacology: told alcohol-given alcohol (A-A, pharmacology + expectancy); told alcohol-given placebo (A-P, expectancy only); and told placebo-given placebo (P-P, control). High AS student participants were randomly assigned to each condition. Following alcohol consumption and absorption, participants underwent a 3 minute voluntary hyperventilation challenge. Preliminary analyses indicate that A-A participants show greater reductions in affective, cognitive and somatic symptoms in response to challenge than A-P and P-P participants. However, A-P participants do not show reduced affective and cognitive symptoms to challenge compared to P-P participants. In fact, correlational analyses suggest a positive relationship between subjective intoxication measures and levels of affective reactivity within the the A-P group (a "reverse expectancy" effect), and a negative relationship between these measures in the A-A group. Implications of these results for prevention of alcohol abuse in high AS individuals are discussed.

Role of Sulfated Cholecystokinin Octapeptide (CCK-8S) in the Modulation of Separation-Induced Ultrasonic Vocalizations in 12 Day Old Rat Pups

Paul D. Mendella, Anna Maria Szczesniak, Richard E. Brown

Rat pups emit ultrasonic vocalizations (UV's) when separated from the dam and littermates. Anxiogenic drugs increase UVs, while anxiolytic drugs decrease UVs. The neuropeptide cholecystokinin (CCK) induces anxiety in health human volunteers and in individuals diagnosed with panic disorder. The purpose of the present investigation was to assess the effects of CCK-8S on UVs, locomotor activity, and body temperature in 12 day old rats pups, 15 or 30 minutes following drug treatment. Pups were injected intraperitoneally with either vehicle or CCK-8S (0.5, 1.0, 2.0, 4.0 $\mu\text{g/kg}$). After 15 minutes, male pups emitted more UVs and less turing behaviour than female pups. This is one of the few studies to find sex differences in rat pup UVs. After 30 minutes CCK-8S (4.0 mg/kg) significantly reduced line crossing frequency. In addition, male pups completed fewer head raises compared to female pups 30 minutes following drug treatment. Neither UV rate or body temperature was significantly altered by CCK-8S after 15 or 30 minutes. These findings suggest that CCK-8S is not directly involved in the modulation of anxiety-related behaviour in the separation-induced UV model of anxiety.

Saccharin must be added to the taste cue to condition robust taste preferences

Cathy Forestell and Vin LoLordo

A reverse-order differential conditioning procedure was employed to condition taste preference in rats. During training, one taste solution (CS+) was presented 3 min after a calorific reinforcer (glucose), and another taste solution (CS-) was presented 3 min after water. For half of the animals, both taste solutions always contained saccharin, whereas for the remaining animals, the tastes were mixed in water. Subsequently, two-bottle tests revealed a significant preference for the CS+ relative condition. These results suggest that without the presence of saccharin in the CS solutions, conditioned preferences do not occur.

Simulating oculomotor behavior within a neural field model

S. D. Simpson, R. Klein, T. Trappenberg

Oculomotor programming can be viewed as the process of integrating exogenous (environmental) and endogenous (intentional) signals into a unified motor goal. Just as many experimental researchers recognize the importance of understanding both the bottom-up and top-down influences on oculomotor programming, we believe it is important to recognize that computational modeling of such systems should also be constrained by multiple levels of analysis; specifically by behavioral and neurophysiological data obtained from the area of interest. By extending and modifying a current neural field model of oculomotor programming (Kopecz and Schoner, 1996) we have developed a computational account of both behavioral and neurophysiological data from a variety of saccadic paradigms including target location preparation (Munoz & Dorris, this meeting), antisaccades (Everling et al, this meeting), and distractor manipulations. In the model, environmental and endogenous signals are integrated within a buildup layer allowing for competition among possible oculomotor programs. The activity of this buildup layer serves as input to a burst layer whose activity represents a specific oculomotor command. This poster will describe the model's architecture and demonstrate its ability to simulate saccadic reaction time patterns and neuronal activity of buildup and burst neurons within the Superior Colliculus. Possible enhancements to the model will also be discussed.

